

What Are Statistics in SQL Server?

Statistics are internal data objects that SQL Server uses to estimate the **distribution of values** in one or more columns of a table or index.

They help the **Query Optimizer** decide:

- Which **index** to use
- What **join strategy** to pick
- How many **rows** will likely match a condition
- How to **optimize execution plans**

 Essentially, statistics = “data about your data.”

Example

Suppose you have a table:

```
CREATE TABLE Orders (  
    OrderID INT PRIMARY KEY,  
    CustomerID INT,  
    OrderDate DATE,  
    Amount DECIMAL(10,2)  
);
```

SQL Server automatically creates **statistics** for:

- The **index keys** (like OrderID)
- Sometimes for **non-indexed columns** when used in a WHERE clause

Example:

```
SELECT * FROM Orders WHERE CustomerID = 123;
```

 When you run this query, SQL Server uses statistics on CustomerID to estimate how many rows match 123, which affects whether it scans or seeks the index.

How to View Statistics

You can view statistics for a table with:

-- List all statistics

```
EXEC sp_helpstats 'Orders', 'ALL';
```

Or see details:

```
DBCC SHOW_STATISTICS ('Orders', 'IX_Orders_CustomerID');
```

This shows:

- **Header** (last updated date, rows, density)
 - **Histogram** (distribution of values)
 - **Steps** (sample of data value ranges)
-

How to Update Statistics

Statistics can become **outdated** after large inserts/updates/deletes.

SQL Server may auto-update them, but sometimes you must **manually refresh**.

Command:

```
UPDATE STATISTICS Orders;
```

Or update all tables in a database:

```
EXEC sp_updatestats;
```

If you want to specify a sampling rate:

```
UPDATE STATISTICS Orders (IX_Orders_CustomerID) WITH FULLSCAN;
```

Why Update Statistics?

Because **stale statistics = bad query plans**.

Example:

- If your table grows from 1,000 to 1,000,000 rows, but statistics weren't updated, the optimizer might still think there are only 1,000 → chooses wrong plan → slow performance.

Auto Create / Auto Update Statistics

You can check or enable these at the database level:

```
ALTER DATABASE YourDB
```

```
SET AUTO_CREATE_STATISTICS ON;
```

```
ALTER DATABASE YourDB
```

```
SET AUTO_UPDATE_STATISTICS ON;
```

SQL Server then automatically manages them for you in most cases.

Summary

Purpose Description

What Metadata about column value distribution

Why Helps SQL Server optimizer choose efficient query plans

Auto Can be automatically created and updated

Manual Use UPDATE STATISTICS or sp_updatestats when needed

View DBCC SHOW_STATISTICS